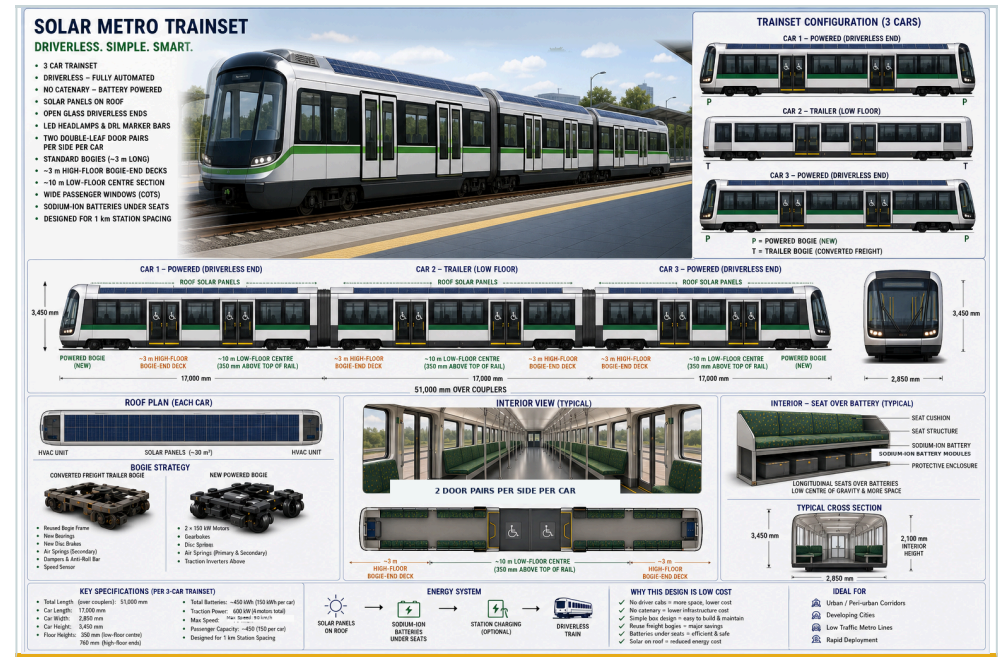


OPEN URBAN RAIL · DESIGN AUTHORITY · LOCAL CAPABILITY

# Build the railway. Keep the knowledge.

OpenSourceRail is an open reference platform for designing, testing, building and operating light metro and urban metro systems.

It is aimed at public owners, domestic engineering firms and qualified suppliers in capital-constrained, high-insolation markets. The public portfolio contains 265 developing-world city models across 43 country programmes. Generated values are planning sensitivities, not bids or funding commitments.



265

developing-world city models across 43 country programmes

~75%

modeled domestic value across the public portfolio

\$900k

local factory-gate LM3 planning target for a 3-car trainset

120

traceable LM3 product rows with visible release gates

## WHY THIS ARCHITECTURE

# A connected system of systems, with narrow authority boundaries.

### COMMODITY HARDWARE

Linux SBCs, deterministic Ethernet and standard radio layers keep value in transferable software and integration.

### CATENARY-FREE ENERGY

The reference uses onboard LFP batteries with station and depot PV; site energy and charging remain deployment studies.

### MACHINE-CHECKABLE SAFETY

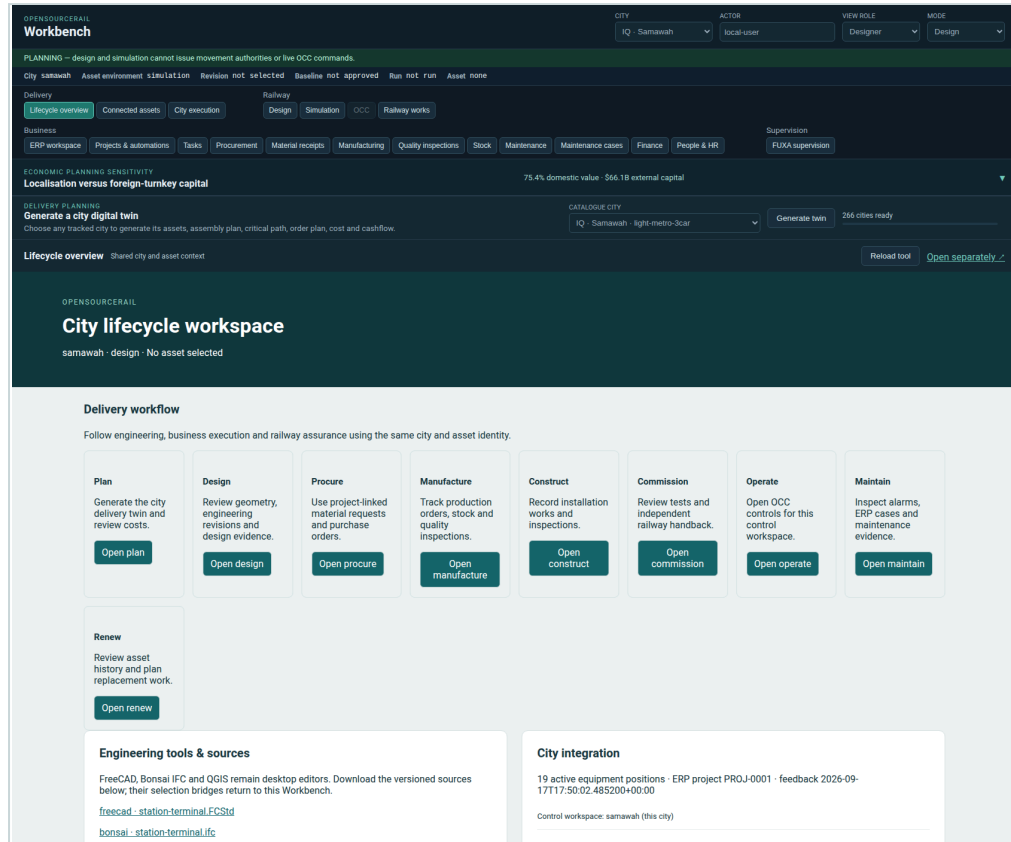
GSN-style claims link formal models, property tests and code while independent assessors retain approval authority.

## ONE DEPLOYMENT PIPELINE

- |    |                                                                |    |                                                                |
|----|----------------------------------------------------------------|----|----------------------------------------------------------------|
| 01 | Choose a city record and country assumptions.                  | 02 | Generate a deterministic city digital twin.                    |
| 03 | Review network, service, cost, energy and fleet outputs.       | 04 | Export civil alignment, CAD, IFC and quantities.               |
| 05 | Build pilot hardware through a COTS/SBC or custom-board track. | 06 | Close survey, supplier, workshop, bench and assessor evidence. |

THE CONNECTED WORKBENCH

# Design, delivery and assurance in one revisioned context.



City, actor, immutable revision, approved baseline, simulation run and selected asset travel together without merging authority boundaries.

## ONE SHARED PROJECT TWIN

City Studio, simulation, OCC training, ERPNext/Frappe HR, FUXA and railway assurance share identity and revision context. Planning and training cannot issue live OCC actions; live mode does not expose design or simulation controls.

## CITY STUDIO

Offline layered GIS authoring, service plans, demand screens, civil choices, deterministic candidate hashes and Git-reviewable immutable revisions.

## DELIVERY CONTROLS

Allowlisted generation produces assets, family-scoped assembly tasks, finite-resource CPM, schedule of values, order-by plans and local/imported cashflow.

## OPERATIONS

Asset registers, work orders, QA hold points, maintenance schedules, defects/NCR, authenticated roles and sealed audit records.

## EVIDENCE

GIS, OSR-ALN, IFC4.3, IDS/BCF, CAD, cost and simulation files remain tied to source and tool hashes. Revision identity proves provenance, not approval.

## START WITH THE LOWER-RISK WEDGE

The first adoptable product is the non-safety owner-operator stack for an existing depot, workshop or pilot corridor: ERPNext + Frappe HR + railway Ops Core + simulator + asset/evidence portal.

|                          |                                 |                                                    |                    |
|--------------------------|---------------------------------|----------------------------------------------------|--------------------|
| Assets                   | QA                              | Work                                               | Handback           |
| stable ids and ownership | OpenSourceRail inspections, NCR | disposition, span-rail maintenance platform orders | evidence and audit |

## EVIDENCE STATUS, STATED PLAINLY

| AREA                           | REPOSITORY STATUS             |
|--------------------------------|-------------------------------|
| Movement authority             | Implemented + tested          |
| Consensus log                  | Implemented + modeled         |
| Obstacle / intrusion detection | Implemented + simulated       |
| Hardware safety nets           | Specified; pilot records next |
| Certification pack             | Pre-submission scaffold       |

## BOUNDARY AND NEXT STEP

This repository is planning and engineering-screening evidence, not a safety certificate, construction release, supplier quote, approval or operating authority. The target is GoA 4, but train control stays in simulation or shadow mode until deployment evidence is independently assessed.

[github.com/moderneecotech/OpenSourceRail](https://github.com/moderneecotech/OpenSourceRail)